

Day 89 A/B: Baseline and Deterministic Stress Fixture

VSEPR-SIM Work Order Consolidation Authority

Day 89

Consolidated scope

This record combines Day 89-A, the measured baseline, and Day 89-B, the deterministic large-particle fixture. The original A and B authorities remain retained as detailed source records; this document is the concise completion boundary for both work items.

The shared producer is `build/dual-backend-3d-data.exe`. Its default fixture remains the four-scene baseline. The optional `--stress-particles N` argument appends a deterministic lattice scene without changing the default package or transferring scientific ownership to a renderer.

Measured acceptance

Measure	Accepted evidence
Baseline scenes / rows	4 / 111
Stress scenes / rows	5 / 1,442
Stress-scene rows	1,331
Stress steps / time step	1 / 10 fs
Fresh generation time	44.0298 ms
Stress validator	PASS

The accepted stress command is:

```
build/dual-backend-3d-data.exe --headless --output out/day89_recheck
--stress-particles 1331
```

The generated manifest reports schema `vsepr.dual_backend_3d.v1`, five scenes, 1,442 particles, and positive generation time. R, PDF, and XLSX sidecars were also produced and validated for this run.

Repeatability evidence

Two fresh producer runs produced identical CSV content. The SHA-256 values are:

- `scenes.csv`: 23EF29B8F1904EA5901CBD8D079EFB10C5A537B9D957873B935AE3A4BA083728
- `particles.csv`: 78D2B1549756987378113D4E7E2EB777043A3DF21B7BEA234C7E0CA0E78BBE6C

The manifest timing field is intentionally run-specific, so manifest hashes may differ even when the scientific CSV rows are identical. The artifact validator accepted both packages, including the greater-than-1,000-particle stress requirement.

Completion decision

Day 89-A/B is **COMPLETE** as one baseline-and-fixture gate. Day 89-C remains a separate completed artifact-contract gate and retains its own authority. Renderer consumption, parity, bead packets, exchange accounting, and browser/report readiness remain later Day 89 work and are not promoted by this consolidation.